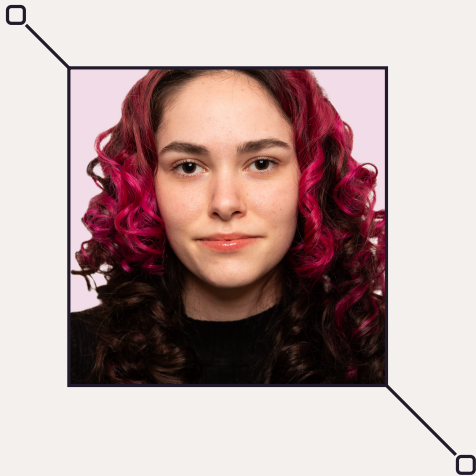




Carla - Maria Băraștean

Automation and Computer Science Graduate

Contact

Email: carlabarastean@gmail.com

Phone: +40 771 732 396

Location: Cluj-Napoca, Romania

Portfolio: website

GitHub: github.com/carlabarastean

LinkedIn: profile

Profile

Automation and Computer Science graduate focused on nonlinear control, model-based development and applied AI.

I build reproducible engineering software, from MATLAB toolboxes and embedded-control models to full-stack applications.

Focus

- Control Systems
- Software Engineering
- Applied AI

Hard skills

Control

MATLAB, Simulink, ASCET, TPT

Programming

Python, C/C++, JavaScript, TypeScript

AI / Vision

TensorFlow, PyTorch, OpenCV

Web / Data

React, Node.js, MongoDB

Tools

Git, Docker, STM32

Soft skills

- Leadership
- Analytical thinking
- Adaptability
- Communication
- Attention to detail

Languages

Romanian (Native)
English (C1)
German (A2)
Italian (A2)

Experience

Model-Based Software Developer

Jul 2024 - Jan 2025

Bosch, Cluj-Napoca

Built and validated a six-component electric-vehicle drivetrain model in Simulink, ASCET and TPT. Generated production C code for STM32 controllers and developed a monitoring interface.

Computer and Network Operator

2019 - Present

ALBALUX COM

Install and configure hardware and networks; diagnose workstation and connectivity issues; provide on-site and remote technical support.

Education

B.Sc. Automation and Computer Science

2022 - 2026

Technical University of Cluj-Napoca

Graduated with a final average of **9.25/10**.

Thesis: *Automation of Nonlinear Control Analysis and Design: A MATLAB Implementation Based on Feedback Linearization.*

Decebal National College

2018 - 2022

Deva, Romania

Mathematics and Computer Science in English

Research

As the Crow Flies, or as the River Flows?

2026

Co-author - manuscript submitted to ICCP 2026

Graph Topology vs. Proximity in Satellite Water-Quality Monitoring.

Controlled comparison of physical river-flow topology and geographic proximity for satellite water-quality monitoring across Germany and Romania.

Selected Projects

EcoSort - Intelligent Waste Recognition

2026

Python, TensorFlow, OpenCV, Arduino

Real-time material classification and optional Arduino-controlled physical sorting.

BudgetLy - Collaborative Finance Platform

2025

React, Node.js, MongoDB, Express

Full-stack platform for financial analytics, recurring transactions and shared-payment tracking.

Selected Activities

CASSINI Hackathon - AquaGraph

2026

3rd in Romania, 1st at the Cluj-Napoca hub

Developed a space-data solution for monitoring and preventing river pollution.

Panel Speaker - IEEE SACI 2025

2025

ADAPTED Research Group

Discussed innovation in STEM and academic entrepreneurship at an international conference workshop.